Juntian Tu

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PROFESSIONAL SUMMARY

Experimental physicist with 5+ years of AMO research experience, specializing in neutral-atom quantum hardware, parallel computing for real-time signal processing, and optical systems. Experience includes GPU-accelerated arbitrary waveform generation (AWG), FPGA development and direct digital synthesis (DDS), high-numerical-aperture (high-NA) optical tweezers, Pound–Drever–Hall (PDH) laser stabilization, and laboratory automation.

TECHNICAL SKILLS

AMO & Quantum Physics: Laser cooling and magneto-optical traps (MOTs); optical tweezers; neutral-atom quantum simulation/computing; spatial-light modulator (SLM) & holography; PDH systems; high-NA optics; Fourier optics

Software: Python, C/C++, CUDA, Mathematica, Linux, Git

Hardware: PCB & embedded systems design, RF systems, AWG, DDS, PID control, FPGA (Verilog), 3D printing, CAD

EDUCATION

Ph.D. Candidate in Physics — Expected: Summer 2028
University of Maryland, College Park

B.S. in Mathematics and Physics with High Honors — Graduation: June 2022
University of California, Santa Barbara — **GPA: 3.93/4.00**

PROFESSIONAL EXPERIENCE

Graduate Research Assistant — August 2022 – Present
UMD/JQI – Trey Porto's Group — *College Park, MD*

- Developed a heterogeneous CPU-GPU–DAC AWG architecture for real-time multi-tone RF control of acousto-optic deflector (AOD) optical arrays; demonstrated control of waveforms containing up to 10,000 tones [1] and real-time tone chirping in experiment.
- Co-developed an FPGA-based DDS system for electronic-sideband PDH locking in Verilog; implemented the laser-locking system and characterized its frequency stability and wide-range tunability while maintaining lock [2].
- Designed a reflective optical system compatible with high-NA optical tweezers to enhance photon collection and enable photon-mediated remote-atom entanglement in future experiments; presented at DAMOP 2026 (poster).
- Developed GPU-accelerated phase-retrieval software for SLM control, enabling rapid phase-pattern updates.
- Evaluated MOT operating parameters through numerical simulation during experiment design. Developed Python-based experimental-control software and embedded RF signal-generation hardware supporting Rb and Yb MOT loading, single-atom optical trapping [3], and real-time atom rearrangement.

Visiting Research Assistant — March 2022 – July 2022
MIT – Ronald Fernando Garcia Ruiz's Group — *Cambridge, MA*

- Applied extreme-value theory to analyze the energy distribution of thermionically emitted electrons; assisted with optical-system assembly and troubleshooting for the experimental apparatus.

Undergraduate Research Assistant — February 2021 – June 2022
UCSB – Mark Sherwin's Group — *Santa Barbara, CA*

- Developed particle-dynamics simulations to investigate time-crystal behavior; helped build optical systems for laboratory experiments.

PUBLICATIONS AND PREPRINTS

1. **Juntian Tu** and Sarthak Subhankar "Fast real-time arbitrary waveform generation using graphic processing units." *IEEE Transactions on Signal Processing*, vol. 73 (2025).
2. (Submitted) **J. Tu**, A. Restelli, et al. "Quadrature amplitude modulation for electronic sideband Pound-Drever-Hall locking." *arXiv* (2024).
3. (Submitted) K. Weber, M. McMaster, M. Anderson, **J. Tu**, et al. "Dual-species alkali and alkaline-earth-like optical tweezer arrays via interferometrically aligned high-NA objectives." *arXiv* (2026).